Anne Kaitlyn Arabelo

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EDUCATION

University of California, San Diego | B.S. in Bioengineering *Expected Dec 2027*

Coursework: Foundation in Biomechanics, Intro to Mechanical Design, Bioinstrumentation, Biomaterials, Control Systems

SKILLS

CAD & Prototyping: SolidWorks, Fusion 360, laser cutting, 3D printing, Arduino and sensors
Software & Data: Python (matplotlib), MATLAB, Excel, statistical validation (t-tests, regression analysis), SQL
Testing & Lab: Size-exclusion chromatography, centrifugation, TEER, ManoView (HRM)
Design Controls & Risk: DFMEA, CGMP, regulatory pathways (PMA, 510(k)) (Coursework)
Training: CITI Human Subjects Research, HIPAA, Laboratory Safety

SENIOR DESIGN CAPSTONE

Quantifying Esophageal Mechanical Work | Senior Design, Mittal Lab *Sep 2025 – Jun 2026*

- Co-authored DFMEA failure modes for HRIM data extraction and diagnostic analysis, defining mitigations (SOPs, calibration checks, outlier detection)
- Processed raw HRIM exports by segmenting the esophageal sphincters (UES/LES), aligning pressure and impedance channels with heatmap QC, and converting impedance to cross-sectional area–tension loops to compute mechanical work
- Analyzed mechanical work in pre-/post-dupilumab HRIM studies of 10 EoE patients, finding significant reductions in 5 of 6 bolus conditions ($p \leq 0.017$) and in the mid-to-distal esophageal body ($p \leq 0.015$)
- Verified impedance-derived cross-sectional area against ultrasound before and after treatment, showing strong agreement

EXPERIENCE

Research Assistant | Mittal Lab (PI: Ravinder Mittal, M.D.), UC San Diego *Jul 2025 – Present*

- Developed a Python tool that processes esophageal motility data from 10 patients and plots EoE disease progression
- Audited 100+ from IRB-approved studies for accuracy and IRB/HIPAA compliance; corrected inconsistencies

Research Assistant, GEAR Program | Aran Lab (PI: Kiana Aran, Ph.D.), UC San Diego *Sep 2024 – Jun 2025*

- Laser-cutted microfluidic blood-brain barrier (BBB) chips from SolidWorks designs for iPSC-derived neural cell culture
- Integrated electrodes into the BBB chip, enabling successful barrier-integrity measurements via TEER
- Isolated red blood cell–derived exosomes using size-exclusion chromatography (SEC) and centrifugation, purifying exosomal RNA to evaluate exosome-mediated RNA delivery across the BBB chips

PRESENTATIONS

- Co-author, "Dupilumab Decreases Esophageal Workload During Bolus Transport in Patients with Eosinophilic Esophagitis (EoE)," *Gastrointestinal Endoscopy* 103:S-684, presented at DDW 2026, Chicago, IL | May 2026
- Presenter, Tan C, Arabelo A, Doshi A, Aran K. "Design and Validation of a BBB-on-Chip Device for Aging and Neurodegenerative Disease Research," GEAR research poster, UC San Diego, La Jolla, CA | May 2025

PROJECTS

Tactical Bio-Core: Portable Probiotic Bioreactor | Biodefense Hackathon, San Diego — **3rd Place** *Oct 2026*

- Designed and built a 3D-printed housing for a portable bioreactor that makes a three-part probiotic for warfighters recovering from GI illness
- Soldered gas and temp/humidity modules for the ESP32 system monitoring external conditions; worked in live demo

Firefighter-Support Robot Dog | Compile Hackathon, UC San Diego — Top 5 finalist *Oct 2026*

- Wired gas and flame sensors so the robot flashes a warning when gas is detected and sends responders its live location
- Pitched the design to judges and showed a live demo of the project on behalf of the team